A rising biocapacity total can mask ecological-capital land drawdown: a two-land model

Composition mechanism · selected simulation · identification limits · separate NFA account evidence

COMPOSITION MECHANISM

annual account contributions; area conserved

FAST-PROVISIONING AREA A_f

annual contribution $b_f A_f$

conversion u_c

ECOLOGICAL-CAPITAL AREA A_c

annual contribution $Y_c(q)A_c$

Reserve A_r: $A_f + A_c + A_r = A_{tot}$; transfers conserve area

ANNUAL ACCOUNT B

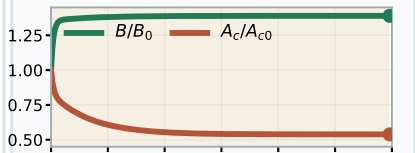
$$B = b_f A_f + Y_c(q) A_c$$

$$\Delta b_{conv} = b_f(D) - Y_c(q) > 0$$

conversion adds to B; A_c falls when $u_c > R_{rc}$

SELECTED FIXED-DEMAND RUN

projected delayed quality; representative —not fitted



+39%	−46.16%	−51.72%
account total	capital area	capital contribution

$E = 0.825$ gha · restoration off · conditional run

IDENTIFICATION & LIMITS

what follows, and under which assumptions

AGGREGATE ≠ ALLOCATION

B alone cannot identify A_c

Independent observations sharpen compatible-area bounds

REDUCED FIXED-DEMAND BRANCH

minimum-area floor at $E_{ceil} = 1.138$ gha

No fold or critical slowing under the stated assumptions

RECOVERY

requires an explicit gated restoration flow R_{rc}

Lower demand alone does not restore A_c

No universal collapse or delay threshold

Model and NFA evidence remain separate

NFA EDITION 2026 · WORLD · 1961–2023 COMPLETE OBSERVATIONS

B +24.86% | B/person -52.71% | cropland 2.8701× | forest 0.8996× | built-up 8.7935×

Annual account values—not physical-stock observations. Estimated 2024–2025 rows excluded; 2022 is a sensitivity check.